

Civil Engineering - 4th Year Technical MCQ Bank

TEXT-ONLY PARSER DEMO COPY | 10 MCQs | 10 marks | 20 minutes

Instructions: Select one option for each question. This copy intentionally uses simple numbered-question and lettered-option text so the PREPAURA document parser can be demonstrated. Answer lines are included for parser validation and are NOT student-facing resource copies.

1. A laterally unbraced compression flange in a steel I-section primarily becomes critical because of which instability mode?

- a. Local crippling
- b. Lateral-torsional buckling
- c. Web bearing failure
- d. Block shear

Answer : b

2. For a steel tension member with staggered bolt holes, the net-section path is generally based on which geometric adjustment?

- a. Adding hole diameters along the critical path
- b. Subtracting hole diameters with stagger allowance for staggered pitch
- c. Ignoring holes because gross area governs
- d. Using only the gross width

Answer : b

3. In the matrix stiffness method, the global stiffness matrix is assembled primarily by transforming and superposing:

- a. Flexibility coefficients of supports only
- b. Element stiffness matrices in global coordinates
- c. Influence lines of members
- d. Plastic hinge rotations

Answer : b

4. For a simply supported beam carrying a moving point load, the influence line for the reaction at the left support is:

- a. Constant 1 everywhere
- b. Linear, decreasing from 1 at the left support to 0 at the right
- c. Parabolic
- d. Zero except at the load point

Answer : b

5. In a conventional activated-sludge process, the main purpose of the secondary clarifier is to:

- a. Oxidize ammonia directly
- b. Separate biological flocs from treated water
- c. Increase raw-water turbidity
- d. Remove coarse grit

Answer : b

6. A wastewater with BOD5 = 200 mg/L and ultimate BOD substantially higher primarily indicates:

- a. No biodegradable organic matter
- b. A biodegradable organic load that continues beyond five days
- c. Only inorganic pollution
- d. Only toxic metals

Answer : b

7. For a horizontal highway curve, superelevation is introduced mainly to:

- a. Increase pavement thickness
- b. Counteract lateral acceleration of vehicles
- c. Reduce stopping sight distance
- d. Increase wheel load

Answer : b

8. If a rigid pavement is designed for repeated wheel loads, the critical structural response is strongly associated with:

- a. Flexural tensile stress in concrete
- b. Only shear stress in soil
- c. Only asphalt rutting
- d. Only thermal conductivity

Answer : a

9. For an idealized two-way slab panel, a direct design consideration when using yield-line theory is the:

- a. Plastic collapse mechanism and corresponding work equation
- b. Laminar flow regime
- c. Darcy friction factor
- d. Buckling load of a column

Answer : a

10. In water chlorination, the contact tank is intended to provide sufficient:

- a. Detention time for disinfection kinetics
- b. Head loss for sediment transport
- c. Aeration for BOD production
- d. Pressure for filtration

Answer : a